



Published in final edited form as:

Brain Inj. 2021 May 12; 35(6): 725–733. doi:10.1080/02699052.2021.1906445.

Early Stage Longitudinal Subcortical Volumetric Changes following Mild Traumatic Brain Injury

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Abstract

Objective: To investigate early brain volumetric changes from acute to 6 months following mTBI in deep gray matter regions and their association with patient 6-month outcome.

Methods: Fifty-six patients with mTBI underwent MRI and behavioural evaluation at acute (<10 days) and approximately 1 and 6 months post-injury. Regional volume changes were investigated in key gray matter regions: thalamus, hippocampus, putamen, caudate, pallidum, and amygdala, and compared with volumes from 34 healthy control subjects. In patients with mTBI, we further assessed associations between longitudinal regional volume changes with patient outcome measures at 6 months including post-concussive symptoms, cognitive performance, and overall satisfaction with life).

Results: Reduction in thalamic and hippocampal volumes was observed at one month among patients with mTBI. Such volume reduction persisted in the thalamus until 6 months. Changes in thalamic volumes also correlated with multiple symptom and functional outcome measures in patients at 6 months.

Conclusion: Our results indicate that the thalamus may be differentially affected among patients with mTBI, resulting in both structural and functional deficits with subsequent post-concussive sequelae and may serve as a biomarker for assessment of efficacy of novel therapeutic interventions.

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Disclosure Statement:

The authors report no conflict of interest. The study was conducted at University of Maryland School of Medicine Center for Innovative Biomedical Resources, Translational Research in Imaging @ Maryland (CTRIM) – Baltimore, Maryland.

Keywords

mild traumatic brain injury; volumetric analysis; post concussive symptoms; Magnetic Resonance Imaging

Introduction

Mild traumatic brain injury (mTBI) accounts for more than 70% of all traumatic brain injury (TBI) (1). The primary injury to the brain is a consequence of sudden acceleration, deceleration, and/or rotational forces exerted on the brain within the skull at the time of the initial impact, often resulting in distributed shearing of axons that is referred to as diffuse or traumatic axonal injury (DAI or TAI respectively). Currently, conventional magnetic resonance (MR) and computed tomography (CT) imaging are often not sensitive in detecting subtle DAI-related injury in patients who experience mTBI; however, the presence of persistent post concussive symptoms suggest that even “mild” injuries may result in damage to the structural integrity and functional communication across the brain (2–4). These persistent symptoms may be due to acute damage associated with the initial injury or may be related to the secondary injury mechanisms that continue to propagate injury across the brain through various biomolecular and cellular pathways, resulting in pervasive degeneration of neurons and glial cells. These secondary injury pathways include but are not limited to chronic inflammation, cellular vascular dysregulation, disruption in plasma membrane and neurotransmitter release, mitochondrial dysfunction, production of reactive oxygen species, altered anaerobic metabolism, and lactic acidosis which together contribute to both necrotic and apoptotic cell death (5,6).

TBI, especially moderate or severe TBI, often leads to substantial cerebral atrophy which can be visualized by a neuroradiologist as a loss of white and gray matter volume and an enlargement of ventricles. This volume loss continues to progress through the chronic stages of injury and affects both cortical and sub-cortical areas (7). Regional volume changes have been shown to correlate with performance on neuropsychological assessments, level of long-term disability, and severity of injury (7–11). While initial research has been conducted in moderate to severe TBI population, recent studies suggests that even milder injuries may result in regional brain volume changes that correlate with persisting post concussive symptoms (12,13). For example, thalamic atrophy was found to be associated with fatigue in veterans with a history of mTBI (14).

Longitudinal volumetric studies in patients with mTBI are, however, limited and results are often inconsistent. Zagorchev et al. reported reduced volumes in the caudate, putamen and thalamus in patients with mTBI at 2 months after injury, but that these volumes appeared to normalize at 1 year after injury; on the other hand, the amygdala and hippocampus showed longitudinal volume loss from 2 month to 1 year post injury, suggesting a potential region specific response to the injury (13). Zhou et al. reported a longitudinal volume loss in the white matter of the bilateral rostral anterior cingulate in 28 patients with post-concussive symptoms from the acute stage of mTBI to 1 year post injury that correlated with anxiety and other post-concussive symptoms (12). Hellstrom et al. reported longitudinal volume

reductions in the left accumbens and right caudate nucleus from 4 weeks to 12 months after mTBI, irrespective of whether or not these patients had the presence of intracranial abnormalities on the day of injury based on CT (i.e., complicated or uncomplicated mTBI) (15). However, they failed to find any association between regional volumes and measures of post-concussive symptoms or global functioning.

Although subcortical atrophy is most commonly reported in patients with mTBI at chronic time points, in vitro studies indicate that primary trauma-induced cell loss may start much earlier post injury, immediately or within hours following injury (16). A more complete understanding of acute structural brain changes following mTBI is critical for early diagnosis and outcome prediction as well as for the initiation of therapeutic treatments or interventions. Therefore, the purpose of this study is to investigate early brain volumetric changes following mTBI from an acute time point of less than 10 days post injury through 6 months post injury with a focus on deep gray matter regions that have been previously implicated in TBI literature. We hypothesize that there are volumetric changes in patients with mTBI over the first 6 months following injury. Furthermore, we anticipate that the degree of regional volume change will associate with severity of post concussive syndrome (PCS) and objective cognitive performance.

Material and Methods

Ethics Statement

The Institutional Review Board of the University of Maryland, Baltimore approved this study. All participants signed the written informed consent with HIPAA compliance.

Participants

Patients presenting with mTBI were prospectively recruited from the R Adam Cowley Shock Trauma Center at the University of Maryland Medical Center. Inclusion criteria for the mTBI group in this study included patients in the age range of 18–70, an admission Glasgow Coma Score (GCS) of 13–15, mechanism of injury consistent with trauma, evidence of physical trauma to the head, and either (a) a positive admission CT or (b) reported loss of consciousness and/or post traumatic amnesia. Exclusion criteria included history of neurological and psychiatric illness, history of prior head injury, stroke, brain tumors, or seizures, and contraindications to MR. Fifty-six patients completed three visits (MRI and behavior assessment), which we refer to as the acute stage (up to 10 days post injury), 1 month (post injury) and 6 month (post injury) (Table 1). Both CT and MRI radiology reads for each patient was obtained from patient electronic medical database. CT imaging findings were based on the admission CT. MRI imaging findings were based on the acute stage conventional MRI sequences (T1, T2, FLAIR, SWI and diffusion weighted images). After reviewing images, we excluded three patients due to motion artifacts in the 3D T1 images, resulting in N = 53 patients that were finally included in this study. As a comparison, we recruited N = 34 neurologically intact healthy volunteers between age of 18 and 70 from local community to serve as the control group. Recruitment of control participants followed the same exclusion criteria as patients. Control participants had one study visit. Both patients and controls were compensated for their time during the visit.

We further subdivided patients into those with post-concussive symptoms (PCS) and those without PCS (no-PCS) based on their self-reported symptoms on the Modified Rivermead Post-Concussion Symptoms Questionnaire (RPQ) (17) at their 6 month visit. In the RPQ, participants rate a series of common post concussive symptoms on a scale of 0-4. The symptoms are categorized as Somatic Symptoms (headache, dizziness, nausea, vertigo, poor coordination, loss of balance), Cognitive Symptoms (trouble concentrating, slow thinking, memory problems, difficulty making decision), Neuropsychological Symptoms (sleep disturbances, depression, anxiety, fatigue, irritability, hyperactivity), and Sensory Symptoms (double blurred vision, sensitivity to light, sensitivity to sound, hearing difficulty, numbness, changes in taste and smell) (18). We calculated the total score on all symptoms as well as total scores from each of the subgroup. We grouped the patients to the PCS and No-PCS groups based on the International Classification of Disease tenth revision (ICD10) symptom criteria at their 6 month visit, where PCS is defined as reporting 3 or more of the following symptoms lasting for greater than three months: headaches, dizziness, sleep disturbances, trouble concentrating, fatigue, memory problems, and irritability at the chronic stage (19).

Participant Assessments—All patients underwent neuropsychological and cognitive assessment at each time point, which included the Military Acute Concussion Evaluation (MACE) (20), the Automated Neuropsychological Assessment Metrics (ANAM) battery, the Glasgow Outcome Scale Extended (GOSE) (24) and the Satisfaction With Life Scale (SWLS) (25,26). For MACE, GOSE, SWLS tests, we calculated the total score as part of the outcome assessments. For ANAM, we also considered the following subtests that assesses various cognitive domains including memory, attention, and processing speed: the code substitution (CS), code substitution delayed (CSD), match to sample (MTS), math processing (MATH), procedural reaction time (PRT), simple reaction time (SRT), simple reaction time repeat (SRT2) (21). For each individual subtest, we extracted a throughput score that incorporates both accuracy and reaction time information (22). We also calculated a weighted throughput score (WT-TH), that has also been referred to as an Index of Cognitive Efficiency (23):

$$\text{WT-TH} = (\text{CS} \times 4.35 + \text{CSD} \times 5 + \text{MTS} \times 6.63 + \text{MATH} \times 8.37 + \text{PRT} \times 2.18 + \text{SRT} + \text{SRT2}) / 7.$$

For all patient condition assessments, we took special care to ensure patient's reported symptoms and disabilities that were specifically attributable to their recent TBI event. Control participants underwent the MACE, ANAM and SWLS tests during their study visit.

MR Data Acquisition—We obtained all MR images on a Siemens Tim Trio 3T MRI Scanner. The imaging protocol included a high-resolution three-dimensional T1-weighted scan using the magnetization prepared rapid acquisition of gradient echoes (MPRAGE) sequence. Due to an imaging protocol update in the middle of the study, the MPRAGE sequence had two variations: (1) using 12 channel receiver head coil, TE = 3.44 ms, TR = 2250 ms, TI = 900 ms, flip angle = 9°, FOV = 230 mm, resolution = $0.9 \times 0.9 \times 2 \text{ mm}^3$, and 72 axial slices parallel to anterior and posterior commissure points (AC-PC)); (2) using 32 channel receiver head coil, TE = 2.91 ms, TR = 2300 ms, TI = 900 ms, flip angle = 9°, FOV = 256 mm, resolution = $1 \times 1 \times 1 \text{ mm}^3$, and 196 sagittal slices to cover the whole head. Other conventional MRI performed included T2 weighted (TE = 83 ms, TR = 5690 ms, FOV

= 230 mm, resolution = $0.6 \times 0.4 \times 5 \text{ mm}^3$, and 26 axial slices with 20% gap), FLAIR (TE = 77 ms, TR = 8000 ms, TI = 2500 ms, FOV = 230 mm, resolution = $1.2 \times 0.9 \times 5 \text{ mm}^3$, and 26 axial slices with 20% gap), and susceptibility-weighted imaging (TE = 20 ms, TR = 28 ms, flip angle = 15° , FOV = 230 mm, resolution = $0.8 \times 0.7 \times 2 \text{ mm}^3$, and 72 axial slices).

Volumetric Analysis—We used the FreeSurfer image analysis suite (version 5.3; <https://surfer.nmr.mgh.harvard.edu>) for image pre-processing and automated subcortical volume analysis. Briefly, the pre-processing procedures included motion correction, skull stripping, Talairach registration, volumetric labeling, intensity normalization, and nonlinear volumetric alignment to MNI305 atlas. Automatic subcortical segmentation based upon a probabilistic structure atlas labelled subcortical tissue into 40 sub-regions. Following inspection of the segmentation results for each participant to ensure the accuracy of the skull stripping and subcortical segmentation, we converted the segmented sub-regions into volume measurements in mm^3 with the *asegstats2table* script. The regions examined in this analysis included thalamus, putamen, caudate, pallidum, amygdala, and hippocampus in the left and right hemispheres. We chose these regions because they have been previously implicated as vulnerable regions following TBI (3,10,12,27). We also extracted the total intracranial volume for each subject for volume normalization.

We used the FreeSurfer longitudinal pipeline (28) to extract reliable volume estimates in patients, with the first imaging time point as the reference. In this pipeline, an unbiased within-subject template space and image (29) is created using robust, inverse consistent registration (30). Several processing steps, such as skull stripping, Talairach transforms, atlas registration, and spherical surface maps and parcellations are then initialized with common information from the within-subject template, significantly increasing reliability and statistical power.

To ensure proper longitudinal analysis, for those participants who had inconsistent T1-MPRAGE acquisitions, we only included data with the same initial T1-MPRAGE acquisition in the analysis. For example, if a participant has sequence type 0 for V1 and V2 and Sequence type 1 for V3, then only V1 and V2 are included in the analysis. This resulted in a total of N=53 cases included in the acute visit, N = 51 cases included in the 1 month visit, and N = 42 cases in the 6 month visit.

We used the following equation to calculate longitudinal volume changes, $(V_n - V_1)/V_1 \times 100\%$, where V_1 is the regional volume at the acute stage, and V_n is the regional volume either at 1 month or 6 month.

Statistical Analysis—We performed all statistical analysis in R (R Core Team, 2018). For all demographic information (mTBI vs. control and PCS vs. No-PCS) comparison, we used the two-sample t-test for continuous variables and chi-square tests for categorical variables. For neuropsychological assessments (MACE, ANAM, SWLS, GOSE) comparison between groups, we used ANCOVA adjusting for age, sex and education.

Following extraction of all regional volumes independently from left and right hemispheres, we tested for laterality in patients using linear mixed model with regional volume as a

dependent variable, visit (acute, 1 month and 6 month), side (left or right), and interaction between visit and side as fixed effects, and participant as a random effect. For regions that had no significant interaction between visit and side, we summed bilateral regions to form a single whole region volume .

Regional volumes from control subjects were compared to patients with mTBI at the acute visit only. We then performed longitudinal analysis in patients with mTBI. We further compared between the PCS and No-PCS mTBI groups at each visit independently. We used ANCOVA for all group comparisons, adjusting for age, sex, sequence type, and intracranial volume.

For the longitudinal analysis of regional volume changes within patients, or subgroups of PCS or No-PCS patients, we used a linear mixed model with visit (acute, 1 month, 6 month) as the fixed effect, participant as the random effect, and age, sex and sequence type as covariates. We included sequence type as a covariate to account for potential systematic bias in using anisotropic voxel size vs an isotropic voxel size (31). Missing data were handled through listwise deletion followed by maximum likelihood estimation of model parameters. For regions with a significant visit effect, we conducted post-hoc Tukey tests for pair-wise comparison between different time points.

Associations between the regional volume changes and various outcome measures were assessed in all patients using partial correlation adjusted for age, gender and sequence type. To correct for multiple comparisons, we pooled all p-values for correlations and used the False Discovery Rate (FDR) method. The significance level was defined as $p < 0.05$.

Results

Participants

Participant demographics are shown in Table 1. There were no significant differences between control subjects and patients with mTBI, or between the PCS and No-PCS subgroups of patients in terms of age, gender, and years of education. There were also no significant differences in days post injury between the two patient sub-groups at any visit. 34%–36% of all patients showed evidence of intracranial injury based on CT and/or conventional MRI at the acute stage post injury. Although there appear to be a higher percentage of patients with either a positive CT or MRI in the PCS group, the difference was not statistically significant.

Participant Assessment

Compared to controls, patients with mTBI had significantly lower MACE total score at both acute and 1 month time points, with recovery at 6 months. Patients also had significantly lower ANAM WT-TH scores than controls acutely, with recovery at 1 month and 6 months. There was no difference in SWLS score between controls and mTBI at any time points. Within the patients, patients with PCS had significantly lower ANAM score than patients with No-PCS at the acute stage. While the scores remained lower in patients with PCS, the difference was no longer statistically significant at 1 month or 6 month visits. On the other hand, patients with PCS had significantly lower GOSE than patients with No-PCS at both

the 1 month and 6 month time points. The two patient groups showed no difference in MACE total scores.

Table 2 summarizes the average patient symptom rating scores for patients in the PCS and No-PCS groups in each of the four categories of symptoms, as well as the total percentage of all patients with symptoms. While almost all patients had symptoms initially, over 60% of all patients still had somatic, cognitive and neuropsychic symptoms at 6 months.

Sub-cortical Volumetric Changes

Laterality in Regional Changes in mTBI—An analysis of the bilateral volumes revealed that there were no significant interactions between “Visit” and “Side” in any of the regions (Visit x Side interaction: Thalamus: $p = 0.74$, Caudate: $p = 0.61$, Putamen: $p = 0.79$, Pallidum: $p = 0.94$, Hippocampus: $p = 0.55$, Amygdala: $p = 0.97$, Accumbens: $p = 0.52$). Due to the fact that both the left and right regional volumes have highly similar patterns of longitudinal changes, we only considered the summation of bilateral volumes of each region for all further analysis. Absolute volumes in bilateral regions are shown in Supplementary Table 1.

mTBI vs. Control—Table 3 summarizes the regional volumes for the patient and control groups at the acute stage. No significant differences in regional volumes were observed between the patients and the control subjects at this acute stage.

Longitudinal Analysis—Among the patients with mTBI, we observed significant longitudinal volume changes in the thalamus and hippocampus. Specifically, both the thalamus and hippocampus had significant volume reductions at 1 month (Table 4). Only thalamic volume remained significantly lower at 6 months. When further divided into PCS and No-PCS groups, only the PCS group showed the same pattern of longitudinal changes in the thalamus and hippocampus (Table 4), while the No-PCS group showed no longitudinal changes in any regions (Supplementary Table 2).

Correlations Between Volume Changes and Outcome Measures

Volume changes from acute to 1 month and from acute to 6 month visits for different regions, and their association with patient symptoms, cognitive performance, and overall outcome (GOSE) at 6 months were investigated.

Volume changes at 1 month—Even though most significant volume changes were observed from acute to 1 month, we found no significant correlations between the 1 month volume change and any of the 6 month outcome measures.

Volume changes at 6 months—Table 5 shows the pair-wise correlations between the 6 month volume change in each of the regions and patient outcome measures at 6 months (symptoms, cognitive performance and overall outcome), adjusted for age, sex and sequence type (FDR corrected). Six-month volume change in the thalamus correlated significantly with many outcome measures: SWLS total score ($r = 0.55$, $p = 0.007$), GOSE score ($r = 0.560$, $p = 0.005$) and the RPQ total score ($r = -0.49$, $p = 0.016$). This suggests that patients

with more thalamic volume reduction tend to have higher symptom ratings high RPG score) and poorer outcomes (lower SWLS, GOSE scores). More specifically, thalamic volume changes significantly correlated with RPQ somatic symptom total score ($r = -0.55$, $p = 0.0053$) and RPQ cognitive symptom total score ($r = -0.47$, $p = 0.022$). Hippocampal volume changes significantly correlated with the GOSE score ($r = 0.56$, $p = 0.008$), RPQ total score ($r = -0.49$, $p = 0.04$), and RPQ somatic symptom total score ($r = -0.43$, $p = 0.04$). Amygdala volume change significantly correlated with SWLS total score ($r = 0.50$, $p = 0.014$), GOSE score ($r = 0.49$, $p = 0.013$) and RPQ somatic symptom total score ($r = -0.45$, $p = 0.024$). Putamen volume change significantly correlated with the GOSE score ($r = 0.56$, $p = 0.029$). No correlations were found between volume changes in pallidum or caudate and any of the behavioral measures. Furthermore, we found no correlations between regional volume changes and MACE or ANAM scores.

Among all regions, the thalamus showed the strongest correlation with numerous outcome measures. Figure 1 shows the scatter plot of 6 month thalamic volume change with different outcome measures that had significant associations in the No-PCS and PCS group.

DISCUSSION

In this longitudinal study of regional volumetric changes over a short period of 6 months post mTBI, we observed significant volume reduction in several subcortical regions including the thalamus and hippocampus. Our study is the first of its kind to report volumetric changes in patients with mTBI as early as one month post injury. The volumetric reduction in the thalamus and hippocampus appears to be the largest at the 1 month time point, with region specific recoveries at 6 months. Across all patients, regional volume change over 6 months in the thalamus, hippocampus, amygdala, and putamen were correlated with patient global outcome and self-reported symptom ratings at 6 months, with larger volume reduction associated with worse outcome. This suggests that subcortical brain regions are differentially affected by mTBI, and the degree of structural changes due to secondary injury mechanisms may be related to the persistence of post-concussive symptoms and poor overall patient outcome.

Cortical and subcortical atrophy have been a consistent finding following moderate and severe TBI. The involvement of these vulnerable regions, including the thalamus and hippocampus, has been reported in other studies of moderate to severe TBI. For instance, volume loss within the hippocampus, a fundamental limbic system structure, has frequently been reported in moderate to severe TBI (27,32,33). Hippocampal changes have also been associated with memory impairment and major depression in neuropsychiatric and neurodegenerative disorders such as Alzheimer's disease (34) and mild cognitive impairment (35). On the other hand, the role of the thalamus as a central relay station in mTBI has been extensively described in work by Grossman and Inglese (36) who suggest that thalamic damage may play a significant role in TBI impairment. Tate et al. noted atrophy of the thalamus in a population of military patients with mTBI who presented with cognitive complaints imaged at 3–24 months post injury compared to an orthopedic control population (37). Furthermore, our group has previously reported alterations in thalamic functional connectivity in the acute stage among PCS patients following mTBI as well as

metabolic changes observed from magnetic resonance spectroscopy (MRS) that correlated with self-reported sensory symptoms compared to those without PCS (38). Microstructural changes in the thalamus that correlated with cognitive recovery in patients with mTBI have also been reported (3).

In this study, we found a significant volume reduction from acute to one month following mTBI in the hippocampus and the thalamus. While the volume reduction normalized in the hippocampus, thalamic volume remained significantly low at 6 months. Similar findings was reported in another longitudinal study of a group of 44 patients with mTBI imaged at 2 months and 1 year post injury, where Zagorchev et al. reported significantly smaller volumes in mTBI than controls in caudate, putamen, and the thalamus at 2 months after injury, which also showed a normalization at 1 year after injury (13). The authors also reported more pronounced differences in amygdala and hippocampus at 1 year, suggesting regionally specific responses to injury. The underlying mechanisms that lead to such volume reduction remains unclear. Given that we did not find volume differences in patients vs. control at the acute time point, we would expect that even if there was acute edema immediately following mTBI, it would have resolved by 10 days. On the other hand, while report of volume atrophy in patients with moderate and severe TBI are more consistent, the discrepancy in region-specific findings in more subtle mTBI likely reflects variable recovery trajectories that may be influenced by the general heterogeneous nature of the injury.

Our data do support a significant relationship between regional volume and patient outcomes at 6 months, especially within the thalamus. It is interesting to note that regional volume reductions appeared to be more associated with self-reported measures as opposed to objective measure of cognitive performance assessed in this study such as the MACE and ANAM. This distinction between self-reported cognitive complaints and cognitive performance measures is supported by recent work that failed to detect associations between the two (but noted associations between various symptoms clusters) and concluded that each represents a unique outcome following mTBI (39).

Among all symptoms, persistent somatic symptoms showed the strongest correlation with 6 month regional volume reductions within the thalamus, hippocampus, and amygdala. The symptoms included in the somatic symptom category include headache and vestibular symptoms (dizziness, nausea, vertigo, poor coordination, and loss of balance). Traditionally headache and vestibular symptoms are most common immediately following injury and diminish over time so it is intriguing that patients with reduced subcortical volumes have more persistent somatic symptoms. This may suggest that persistent headache and vestibular symptoms are indicative of a greater severity of initial injury or a slower trajectory of functional recovery; however, additional work using more objective vestibular assessments is needed to further tease apart this association. In addition, it is likely that variable progression and recovery of regional volume and symptom presentation continue past the 6 month time point and that measurements taken at more chronic time points may demonstrate different associations between regional volume loss and symptom categories.

Since the sequelae of mTBI can be quite variable, it is likely that certain patients may suffer from symptoms for prolonged periods of time or become predisposed to other conditions.

For instance, in addition to demonstrating associations between the volume of limbic structures such as the hippocampus and amygdala and PCS symptoms and quality of life measures following mTBI in this analysis, structural and functional disruption of limbic structures has also been frequently implicated in other disorders such as posttraumatic stress disorder (PTSD). There is substantial overlap between symptoms associated with PCS and those associated with PTSD, and individuals with co-morbid PTSD and mTBI frequently present with more severe symptoms than those of PTSD or TBI alone (40–42). However, there is conflicting evidence regarding the specific contributions of mTBI or PTSD to persistent symptoms (43,44). The findings presented in this analysis lend evidence to theories that structural and/or functional damage to limbic circuits that occur due to mTBI may predispose individuals to experience PTSD following a traumatic event. But, as this current project did not directly assess PTSD symptomology, future work is needed to further investigate this conjecture.

One limitation of the study includes the limited sample size, and inclusion of only one control visit, which could have resulted in sample bias contributing to the differences noted between control and patients with mTBI. It is also possible that medical comorbidities, such as hypertension, cardiovascular disease, or diabetes, may also affect brain volumes, although such information was not obtained and accounted for in this study. However, the longitudinal analysis allows more robust inferences with limited sample size. Furthermore, as mentioned earlier in this discussion, the chronic time point used in this study is only 6 months post injury which is still a relatively early time, and it is likely that recovery and progression may continue to happen at this time point. Therefore, additional work following cortical and sub-cortical atrophy in patients with mTBI through a longer period of time is needed. Furthermore, there is an imaging protocol change in the middle of the study which is not ideal. The initial structural imaging protocol did not contain isotropic voxels. While voxel size does not affect the reliability in extracted volumetric measurements as suggested in (31), there could be a systematic bias in the measurement. The effect of anisotropic voxels and isotropic voxels should be minimum given that we have used the sequence type as a covariate in all correlations, and we conducted all longitudinal analysis on datasets with consistent imaging protocols only. Finally, although this analysis specifically focused on longitudinal changes in sub-cortical volumes, additional work investigating the structural and functional connections of these regions as well as regional volume changes of additional brain regions is needed to continue to develop a greater understanding of this heterogeneous injury. In the future, we plan to extend our study to larger populations over longer durations using an isotropic resolution imaging protocol to confirm our findings from this preliminary study.

CONCLUSION

The present work demonstrates regional volume loss in multiple subcortical regions (thalamus and hippocampus) in patients with mTBI at 1 month following injury, with thalamic volume loss persisting at 6 months. The results suggest that these regional changes are greater in patients who report persistent symptoms and that there appears to be an association between the degree of regional volume loss and severity of self-reported symptoms. The fact that regional volume change within the first 6 months is associated with

patient symptom recovery supports the potential for the use of early interventions during this period to improve long term patient outcome and quality of life.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

Acknowledgments

The work is supported by NIH under grant 5R01NS105503; DOD under award W81XWH-08-1-0725. The views expressed in this manuscript are those of the authors and do not reflect the official policy of the Department of the Army/Navy/Air Force, Department of Defense, or U.S. Government.

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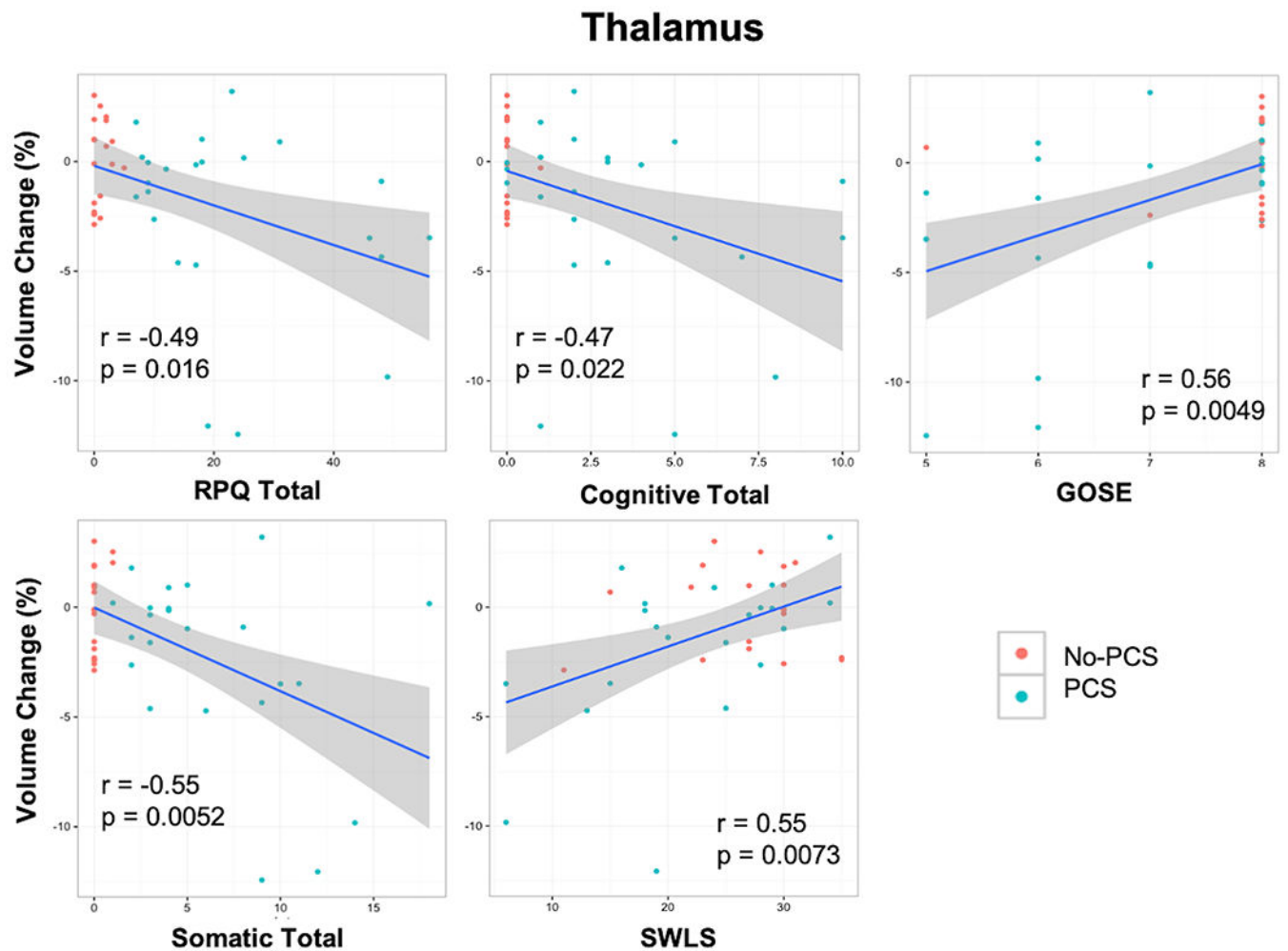


Figure 1.

Correlations between 6 month thalamus volume changes (from acute stage) and 6 month outcome measures in patients with mTBI. Only significant correlation with $p < 0.05$ (FDR corrected) were shown here. Shaded area shows 90% confidence interval. The overall multiple linear regression model correlation coefficient and the corresponding p-values are shown for each association.

Table 1.

Participant demographic information and clinical indication for control and patients with mTBI at different visits (acute: < 10days, 1 month, 6 months). Patients were further grouped to PCS and No-PCS based on their presence of PCS at 6 months. Positive MR/CT indicate visible intracranial abnormalities at initial visit based on MR or CT. MACE: Military Acute Concussion Evaluation total score. ANAM: Automated Neuropsychological Assessment Metrics weighted through-put score. SWLS: Satisfaction With Life Scale total score. GOSE: Extended Glasgow Outcome Scale. Values shown are Mean $\pm$ SD. Group differences were assessed with independent two sample t-test for demographics and Chi-square tests for gender and positive CT or MR.

	Control	mTBI	p-value (Control vs. mTBI)	No-PCS mTBI	PCS mTBI	p-value (PCS vs. No-PCS mTBI)
N	34	53		23	30	
Age (yrs)	39 $\pm$ 18	43.1 $\pm$ 16.0	0.27	38.6 $\pm$ 15.7	46.6 $\pm$ 15.6	0.07
Gender (Male)	18 (53%)	38 (72%)	0.54	19 (83%)	19 (63%)	0.12
Education (yrs)	15.2 $\pm$ 2.3	14.4 $\pm$ 2.6	0.14	14.9 $\pm$ 2.8	13.9 $\pm$ 2.4	0.18
Positive CT	NA	19 (36%)		7 (30%)	10 (33%)	0.57
Positive MR	NA	18 (34%)		6 (26%)	10 (33%)	0.37
Days post injury						
Acute	NA	6.8 $\pm$ 3.3		6.6 $\pm$ 3.0	6.9 $\pm$ 3.6	0.69
1 month	NA	37.4 $\pm$ 10.6		37.1 $\pm$ 13.6	35.3 $\pm$ 6.0	0.56
6 month	NA	201.9 $\pm$ 28.9		204.8 $\pm$ 25.3	195.9 $\pm$ 30.9	0.25
MACE^{\$}	27.9 $\pm$ 1.9					
Acute		25.8 $\pm$ 3.0	0.0056*	25.8 $\pm$ 3.2	25.9 $\pm$ 2.9	0.93
1 month		26.6 $\pm$ 2.2	0.036*	27.5 $\pm$ 2.2	26.2 $\pm$ 2.0	0.13
6 month		27.1 $\pm$ 2.2	0.23	27.5 $\pm$ 2.0	26.9 $\pm$ 2.4	0.51
ANAM^{\$}	218.4 $\pm$ 43.5					
Acute		187.9 $\pm$ 42.3	0.023*	211.1 $\pm$ 28.2	167.8 $\pm$ 42.9	0.032*
1 month		199.1 $\pm$ 41.6	0.16	217.6 $\pm$ 37.0	189.2 $\pm$ 40.5	0.091
6 month		203.7 $\pm$ 43.3	0.29	222.0 $\pm$ 37.3	192.5 $\pm$ 43.5	0.081
SWLS^{\$}	26.5 $\pm$ 6.7					
Acute		25.1 $\pm$ 7.7	0.51	25.1 $\pm$ 7.5	25.2 $\pm$ 8.0	0.56
1 month		25.3 $\pm$ 7.2	0.55	25.9 $\pm$ 6.5	24.8 $\pm$ 7.8	0.38
6 month		24.4 $\pm$ 7.8	0.32	26.0 $\pm$ 7.6	22.5 $\pm$ 8.1	0.1
GOSE^{\$}						
Acute	NA	5.89 $\pm$ 1.67		6.5 $\pm$ 1.7	5.4 $\pm$ 1.5	0.13
1 month	NA	6.72 $\pm$ 1.30		7.4 $\pm$ 1.0	6.2 $\pm$ 1.3	0.0009*
6 month	NA	7.23 $\pm$ 1.11		7.8 $\pm$ 0.7	6.8 $\pm$ 1.2	0.0044*

^{\$}: tests were adjusted for age and gender.

* indicate significant differences $p < 0.05$.

Table 2.

Participant symptom rating scores at different visits. RPQ total is the Rivermead Post Concussion Symptoms Questionnaire total score. Somatic, cognitive, neuropsychological and sensory total scores are the total scores of subgroups of symptoms. Values shown are mean $\pm$ SD. Also reported are percent patients with symptoms for each symptom category with a score of 1 or greater.

	No-PCS	% with Symptom	PCS	% with Symptom	All mTBI % with Symptom
RPQ Total					
Acute	5.7 $\pm$ 5.3	87.5%	26.6 $\pm$ 19.6	100.0%	94.9%
1 month	6.2 $\pm$ 10.3	57.0%	21 $\pm$ 14.0	100.0%	81.6%
6 month	1.1 $\pm$ 1.4	47.8%	21.4 $\pm$ 15.4	100.0%	77.4%
Somatic Total					
Acute	1.9 $\pm$ 2.5	62.5%	7.5 $\pm$ 6.2	100.0%	84.6%
1 month	2.1 $\pm$ 3.9	42.8%	5.1 $\pm$ 3.7	89.3%	69.4%
6 month	0.3 $\pm$ 0.5	21.7%	5.6 $\pm$ 4.4	90.0%	60.4%
Cognitive Total					
Acute	1.1 $\pm$ 1.7	56.3%	6.0 $\pm$ 4.5	91.3%	76.9%
1 month	1.6 $\pm$ 2.7	42.8%	5.1 $\pm$ 3.7	92.9%	71.4%
6 month	0.3 $\pm$ 0.7	26.1%	5.5 $\pm$ 4.3	96.7%	66.0%
Neuropsych Total					
Acute	1.9 $\pm$ 1.7	75.0%	7.7 $\pm$ 5.6	91.3%	84.6%
1 month	1.8 $\pm$ 3.2	42.8%	6.4 $\pm$ 4.9	82.1%	65.3%
6 month	0.3 $\pm$ 0.6	26.1%	6.3 $\pm$ 4.8	93.3%	64.2%
Sensory Total					
Acute	0.7 $\pm$ 1.3	37.5%	5.4 $\pm$ 5.5	78.3%	61.5%
1 month	0.8 $\pm$ 1.5	28.6%	3.9 $\pm$ 4.1	75.0%	55.1%
6 month	0.1 $\pm$ 0.5	8.7%	4.1 $\pm$ 4.2	73.3%	45.3%

Table 3.

Regional volumes (mm³) in control and mTBI at acute time point. p-value shows group difference with ANCOVA adjusting for age, gender and sequence type and intracranial volume sizes.

Regions	Control		mTBI		p-value (Control vs. mTBI)
	Mean	SD	Mean	SD	
Thalamus	13852	2264	13802	1895	0.85
Caudate	7159	1153	7470	981	0.12
Putamen	12326	1812	12495	1585	0.49
Pallidum	3544	638	3655	678	0.26
Hippocampus	8053	964	7995	811	0.7

Table 4.

Longitudinal changes in regional brain volume (mm³) in patients with mTBI, and specifically in patients with PCS, at three visits. Regions with significant visit effect from linear mixed model analysis were followed with pair-wise comparison using post-hoc Tukey HSD test.

All mTBI								
Regions	acute		1 month		p-value vs. acute	6 month		p-value vs. acute
	Mean	SD	Mean	SD	Mean	SD		
Thalamus	13802	1895	13524	2008	< 0.001 *	13774	1986	0.047
Caudate	7470	981	7472	983		7384	1009	
Putamen	12495	1585	12386	1555		12452	1528	
Pallidum	3655	678	3579	633		3645	669	
Hippocampus	7995	811	7829	842	0.0025 *	7999	865	0.84
Amygdala	3065	478	3068	509		3122	525	
PCS								
Regions	acute		1 month		p-value vs. acute	6 month		p-value vs. acute
	Mean	SD	Mean	SD	Mean	SD		
Thalamus	13667	2133	13311	2277	0.0013	13613	2270	0.013
Caudate	7283	1021	7283	1041		7186	1020	
Putamen	12164	1589	12048	1506		12195	1398	
Pallidum	3512	650	3438	632		3549	662	
Hippocampus	7912	856	7718	943	0.008	7843	838	0.44
Amygdala	3098	535	3086	560		3087	494	

* indicate significant differences $p < 0.05$ between the two groups.

Table 5.

Correlation coefficients and p-values (adjusted for age, sex, seq type) between regional volume changes from acute to 6 month and outcome measures at 6 months.

	SWLS	MACE	GOSE	ANAM	RPQ Total	Somatic Total	Cognitive Total	Neurosycho Total	Sensory Total
Thalamus									
r	0.55[*]	0.27	0.56[*]	0.30	0.49[*]	0.55[*]	0.47[*]	0.42	0.39
p-value	0.007	1.00	0.005	0.72	0.016	0.005	0.022	0.067	0.12
Caudate									
r	0.28	0.37	0.22	0.20	0.22	0.30	0.25	0.22	0.19
p-value	0.88	0.08	0.85	1.00	0.83	0.28	0.54	0.81	1.00
Putamen									
r	0.43	0.33	0.50[*]	0.34	0.42	0.43	0.45	0.43	0.35
p-value	0.17	1.00	0.030	1.00	0.18	0.14	0.09	0.15	0.78
Pallidum									
r	0.26	0.38	0.45	0.36	0.33	0.38	0.30	0.30	0.31
p-value	1.00	0.19	0.06	0.31	0.53	0.21	0.93	0.90	0.85
Hippocampus									
r	0.41	0.25	0.52[*]	0.25	0.43	0.43[*]	0.38	0.42	0.37
p-value	0.06	1.00	0.008	1.00	0.040	0.040	0.12	0.05	0.14
Amygdala									
r	0.50[*]	0.23	0.49[*]	0.27	0.37	0.45[*]	0.34	0.33	0.28
p-value	0.014	1.00	0.013	0.83	0.12	0.024	0.21	0.23	0.54

* indicate significant correlations at $p < 0.05$ (FDR-adjusted).